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The dynamic interaction between oil prices and freight rates in maritime trade: a bootstrap rolling window approach

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Abstract

The study uses the rolling window technique to assess the causal link between the oil price (OP) and the Baltic Dry Index (BDI) from 2000 to 2022. The results support the competitive market hypothesis, suggesting that OP and BDI have a significant relationship. The results indicate that knowledge of potential changes in OP can help to lessen the risk of abrupt changes in BDI. Similarly, information on various sources of OP changes can assist the shipping industry in taking practical measures to minimize the impact of shocks. The stability of global maritime trade is vital to the global economy because its massive shares and abnormal behaviour can negatively affect the oil market and international trade. This helps ensure timely measures are taken to avoid unforeseen events and maintain stability. Therefore, the study is a valuable contribution to the shipping industry stakeholders in terms of information about the specific period with underlying factors.

Keywords Shipping industry, Baltic dry index, Oil prices, Global maritime trade, Granger causality

1 Introduction

Transportation is essential because it facilitates trade and promotes economic growth. Maritime transport occupies an exclusive place in the world economy; 90% of international trade is carried across the oceans [24]. It has become an essential shipper of commodities such as metals, grains, and fossil fuels due to the random distribution of resources, economies of scale, and economic

geography [33]. The Baltic Dry Index (BDI) is a recognized tool for measuring changes in freight costs of various commodities, which may be vulnerable to oil prices (OP) [25]. Changes in BDI can indicate global trade and supply chain shifts, indirectly influencing oil prices. An increase in BDI may signal economic growth, while a rise could indicate economic slowdowns or disruptions impacting the shipping sector and oil demand. The changes in OP remain the leading risk faced by the shipping industry, which directly impacts profitability [14]. OP fluctuations constantly affect shipping freight; a rapid increase will bring devastating effects, while a sudden fall boosts the profit in the short run. The higher OP can add costs to the freight; therefore, price fluctuation is one of the contributing factors. The fuel cost for the shipping industry consists of 50–60% of the operating cost, and rising OP is not a good sign for maritime trade [15].

The shipping industry closely monitors OP trends because of its link with the cost of products shipped from

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producers to consumers. However, the pre-recession period indicates that OP touches its highest level because of strong global economic growth, which creates a higher demand for shipping and ultimately moves up BDI. Moreover, the demand from emerging economies such as China and India increase oil demand and maritime trade, ultimately increasing BDI. Also, the financial depression in 2008 declined the demand for commodities, negatively influencing global maritime trade. Similarly, OP collapsed due to the lowest level of demand caused by the economic downturn [19]. Likewise, the decline in global trade may negatively affect the shipping services demand, which is shown in low BDI. The international oil market shows a mixed trend in the post-recession period, mainly driven by the slowing growth of China, excessive OPEC supply, and increasing shale oil production [18]. Likewise, the global maritime trade continues to show a stable trend, and OP moves synchronously with BDI. The U.S.-China trade war may impact OP and the BDI by decreasing trade volumes, increasing shipping costs, and generating market uncertainty. These factors could result in reduced oil demand, which would subsequently lower prices and affect shipping activity. Moreover, the trade tensions could drive a shift toward alternative energy sources, altering the relationship between oil prices and shipping demand [7]. The oil market collapses due to the sharp global economic depression driven by COVID-19. OP has turned negative for the first time in history, which may significantly impact international trade, tailed by the lowermost commodities prices and falling BDI. The shortage of ships and containers cannot meet the unexpected increase in demand which has affected trade flows. The rapid recovery of China's economy in 2020 sustains the smooth flow of international maritime trade. Furthermore, trade flows are supported by containerized goods exported from China to the US. However, weak oil demand and OPEC production cuts affected container demand in 2020, while the market gradually recovered and supply improved in 2021. The crude oil price hit a high level in 2021 due to economic recovery, and the prospect of the international energy market may increase demand for large crude oil carriers. The need for raw materials drives the shipping industry in the second quarter of 2021, and the freight cost is at the highest level [26]. The highest shipping cost has adversely affected commodity prices in the last quarter of 2021. Meanwhile, the freight cost bubble burst in October 2021, mainly caused by the stability of the supply chains, which may cause a decline in freight costs. The Russia-Ukraine war could significantly impact the relationship between OP and the BDI. Disruptions to shipping routes may increase transportation costs while rising oil prices can elevate operating expenses for shipping companies. Additionally, economic uncertainty might affect global trade volumes

and the demand for shipping services, potentially harming the BDI. Evolving trade patterns and inflationary pressures from rising oil prices further complicate this relationship.

This research aims to evaluate the causal relationship between OP and BDI while taking into account time-varying aspects. Oil is the primary energy to promote global economic development and provides most energy used for transportation worldwide [20]. Likewise, maritime transport comprises more than 80% of international merchandise trade and relies heavily on oil. Thus, OP is an extremely deciding factor in the freight cost, negatively impacting BDI. The study contributes by identifying macro-level relationships and the underlying association between OP and BDI in the subsamples. Moreover, most studies such as Shi et al. [36], Bildirici et al. [8], Yıldız and Bucak [45], Khan et al. [20], and Khan et al. [21] analyzed the impact of OP on the shipping industry and ignore the assessment of the relationship in the other direction. Hence, we analyze the mutual relationship between OP and BDI, and the findings suggest a causal association. It makes shipping a crucial part of oil demand and supply due to its massive volume of international trade. The previous literature about OP and BDI has discussed the causality in the full sample, which may experience structural changes due to several political and economic episodes. The existence of the structural changes can cause biased results, indicating that the link between OP and BDI undergoes time-varying uncertainty. Therefore, the present study considers the time-varying features to estimate causality between OP and BDI through the rolling window. The outcome supports the competitive market model and states that OP and BDI have a significant causal relationship. Moreover, OP variations can be reflected in freight costs, while BDI changes may decrease shipping services demand. Similarly, the knowledge of potential OP may lessen the risk of abrupt changes in BDI. Also, information on various sources of OP changes can help the shipping industry to minimize the impact of shocks. Furthermore, the equilibrium of demand and supply in world marine commerce may be a positive indicator for OP, which is critical to global economic development. Therefore, to maintain stability and smooth operation, the most important is identifying the behaviour of OP and BDI. This ensures timely measures are taken to avoid unforeseen events and maintain stability.

The remainder of this work provides a critical analysis of the existing research, beginning with a description of the bootstrap rolling window approach and then on to a discussion of the competitive market model. A description of the data and an empirical evaluation of the findings follow the discussion and robustness results. The last section provides an overall summary.

2 Theory and literature review

2.1 Theoretical framework

It is argued that the international oil market is directly related to BDI and OP is considered the main risk to the shipping industry (Kavussanos and Visvikis, 2006; [14]), which monitors OP trends carefully [15]. This study employs the competitive market model (CMM) proposed by Beenstock (1985). This model posits that a highly competitive maritime market is influenced by various factors affecting transportation costs, such as OP, global trade volume, economic outlook, and the market for shipping services. It demonstrates how rising international commerce across the seas drives the need for shipping services. [11]. The increase in OP upset shipping services and vice versa. Low oil prices benefit the shipping industry by creating higher profits and lower operating costs [36]. The sudden drop increases the profitability of the shipping industry in the short term. Also, BDI can significantly contribute to the OP because the rising shipping costs in the BDI could disrupt crude oil supply chains, impact maritime infrastructure, and potentially cause bottlenecks in oil transport, which may impact OP. The increase in fuel prices due to higher economic growth can easily be reflected in freight. On the other hand, changes in BDI will slow down the demand for shipping services, creating a break in supply and ultimately affecting OP. The finding indicates a mutual causal association between OP and BDI due to various events.

2.2 Literature review

The existing literature covers the discussion about the OP impact on shipping freight and concludes that oil is one of the leading contributors to freight. In this regard, Notteboom and Vernimmen [29] found that OP fluctuations were extremely harmful to freight returns. Alizadeh and Nomikos [3] showed OP and freight markets have a long-term association. Shi et al. [36] inspected the connection between OP fluctuations and the freight market, and it was found that the low supply of crude oil supply has a negligible influence on BDI. Ruan et al. [34] explore that the movement between BDI and OP is significantly multifractal. Moreover, the correlation was persistently strong in the short term. Sun et al. [42] estimated crude oil derivatives and dry bulk freight rates and found that BDI has a significant impact on OP fluctuations.

They find that BDI dependent on OP fluctuations and future freight agreements were extremely vulnerable to volatility [31]. Mou et al. [27] found a stronger correlation between OP and export cargo flow than the import cargo in Europe, North Asia, and East Asia. The shipping cost decreased after the decline of OP and vice versa. Said and Giouvris [35] showed that BDI negatively correlated with economic growth. Choi and Yoon [12] examined the channel of OP change to shipping freight. The

finding explored conditional dependence patterns and asymmetry. Additionally, the dependence increased with the economic boom in the short run. Açık and Ayaz [1] found that oil is essential to freight income. Similarly, the freight rates are determined by the demand of market conditions.

Moreover, some studies examined OP impact on macroeconomic indicators, which is reflected in BDI. Bakshi et al. [5] and Apergis and Payne [4] analyzed the relationship of the macroeconomic indicators with BDI. They concluded that economic activity, OP and commodity demand change were the major determinants of BDI in the short run. Oomen [30] revealed that BDI was driven by supply–demand and other economic measures like the exchange rate and OP can have an insignificant impact. Bildirici et al. [8] explained the importance of BDI as an economic indicator and its dependency on OP, making BDI vulnerable to global demand and manufactured goods. Bildirici et al. [9], Tsioumas et al. [44] and Yin et al. [46] assessed the influencing factors of BDI, which consisted of different market conditions, routes, gold prices, steel production, shipbuilding cycles, and speculative investments. The findings established a consensus that lists influencing elements have a causal effect and contribute to the freight market. Yıldız and Bucak [45] analyzed the predictive factors of BDI and concluded that OP has a significant positive impact. Su et al. [39, 41] examined the freight market bubble determinants, and the result confirmed that OP was one of the potential explanations for freight market bubbles. Khan et al. [20] explored the causal nexus between OP and BDI in the short run. Similarly, Khan et al. [21] explained that OP was one of the main contributors to BDI while geopolitical uncertainty pushed freight rates.

The literature explained the freight market response regarding crude oil as a maritime trade commodity. For example, a major portion of crude oil is carried by shipping, and fluctuations in demand and supply may affect it [47]. Thus, the study considers oil as the transportation cost of the maritime industry and its impact on BDI. Moreover, prior studies such as Shi et al. [36], Sun et al. [42], Ruan et al. [34] Said and Giouvris [35], and Mou et al. [27] covered the one-sided impact running from OP to BDI due to its primary role of energy source in the global economy. However, the literature ignores that BDI can play a significant causative contribution in OP. Rising shipping costs in the BDI could disrupt crude oil supply chains, impact maritime infrastructure, and potentially cause bottlenecks in oil transport. Consequently, assessing the causal link between OP and BDI and vice versa is extremely important. The study aims to evaluate the causality between OP and BDI. This study allows structural changes to analyze the causal link between OP and BDI across subsamples. Similarly, most studies such as Shi et

al. [36], Bildirici et al. [8], Ruan et al. [34], Mou et al. [27], and Said and Giouvris [35] evaluate the full sample relationship of OP on BDI. However, these methods experience low power to detect time variation, resulting in a biased outcome. By using a rolling window, accounting for time-varying features, and recognizing the structural changes brought on by external shocks, the research can overcome these drawbacks and deliver more precise findings.

3 Methodology

3.1 Bootstrap full-sample causality test

The study uses a rolling window methodology to examine the causal relationship between OP and BDI. This approach is suitable for discovering the causal relationship while considering structural breakdowns. It includes a parameter stability test, whole phase, and rolling sub-sample. The technique involves key model specifications that typically include the length of the rolling window, the stationarity of the time series, the presence of linear relationships between variables, and the absence of structural breaks. However, due to the lack of stationarity, the data series does not exhibit conventional asymptotic properties. The absence of these features renders VAR estimates inappropriate [38]. To address this issue, Toda and Yamamoto [43] suggest the Wald test, which calculates the asymptotic distribution. In addition, Shukur and Mantalos [37] use Monte Carlo simulation to alter the size and power characteristics of the small and medium-sized upgraded Wald. The size and power problem are resolved using a residual-based bootstrap (RB) approach. Consequently, the current research employs the technique to evaluate the causal relationship between OP and BDI. The VAR (p) approach is demonstrated as follows:

$$z_t = \delta_0 + \delta_1 z_{t-1} + \dots + \delta_p z_{t-p} + \varepsilon_t, t = 1, 2, \dots, T \quad (1)$$

where, $\varepsilon_t = (\varepsilon_{1t}, \varepsilon_{2t})$ refers to the white noise technique with zero mean. The appropriate lag duration is determined according to the SIC criteria. The Equation comprises two sub-vectors:

$$\begin{bmatrix} OP_t \\ BDI_t \end{bmatrix} = \begin{bmatrix} \sigma_{10} \\ \sigma_{20} \end{bmatrix} + \begin{bmatrix} \sigma_{11}(L)\sigma_{12}(L) \\ \sigma_{21}(L)\sigma_{22}(L) \end{bmatrix} \begin{bmatrix} OP_t \\ BDI_t \\ GMT_t \end{bmatrix} + \begin{bmatrix} \mu_{1t} \\ \mu_{2t} \end{bmatrix} \quad (2)$$

where $\sigma_{ij}(L) = \sum_{k=1}^{p+1} \sigma_{ij,k} L^k$, $\tau_{ij,k} L^k$, $j = 1, 2$, and L is the lag operator, defined as $L^k x_t = x_{t-k}$. GMT plays a catalyst between OP and BDI, which is used as a controlling variable [26]. Equation (2) checks the influence of OP on BDI and vice versa.

3.2 Parameter stability test

A consensus exists that the parameter of the VAR model within the full sample does not fluctuate. However, structural changes may lead to doubtful characteristics and spurious outcomes [6]. Consequently, several investigations have demonstrated that parameter variability is a problem [16]. The above is resolved through tests of short-term parameter stability. In addition, Andrews and Ploberger [2] provide the *Exp-F*, *Mean-F* and *Sup-F* tests for assessing short-run parameters consistency. Similarly, a test for evaluating the total parameter consistency in the VAR system is presented by Nyblom [28] and Hanson [17]. It examines the stability of the parameter allowing structural cracks using LR statistics. The parametric bootstrap method is also used to compute the critical and p -values.

3.3 Rolling-window test

The rolling window test investigates complete sample causation while avoiding structural variations [6]. Furthermore, the method is suitable for observing the structural shifts and time-varying properties of OP and BDI. Pre-testing is particularly appropriate when the entire dataset is non-stationary and exhibits unpredictability amongst sub-samples. A stationary rolling window with l full-size observations is changed into an order of $T-l$ subsamples of $\tau l + 1, \tau l, \dots, T$ for $\tau = l, l + 1, \dots, T$. The RB-based revised LR test is employed to estimate the subsample's causality. The time-varying characteristics and extent of the causal link between OP and BDI are determined from the bootstrap p -values of the identified LR-statistic rolling by $T-l$ subsamples. The result from OP to BDI is equal to the average of all bootstrap estimates and is proved by the formula $N_b^{-1} \sum_{k=1}^p \sigma_{21,k}^*$, where N_b indicates the number of bootstrap repetitions. Also, the causality of BDI on OP is expressed by the following formula $N_b^{-1} \sum_{k=1}^p \sigma_{12,k}^*$. The values of $\sigma_{21,k}^*$ and $\sigma_{12,k}^*$ in the VAR model are evaluated using Eq. (2). Parameter approximation accuracy and sub-sampling model representation are competing goals. The study uses 24 window size 1,000 bootstrap iterations to ensure robust results. See Fig. 1 for detail step by step procedure.

4 Data

The paper uses monthly data from 2000:01 to 2024:11. The steady global economic growth increases oil consumption, pushing OP in 2000. On the other hand, BDI starts rising, and OP can be one of the explaining factors. However, there is a mismatch between demand and supply due to low production that exerts pressure on OP and culminates in the collapse in 2001, coupled with the 9/11 terrorist attack. It is a reason for the higher uncertainty about the global economic scenario and a trade that has

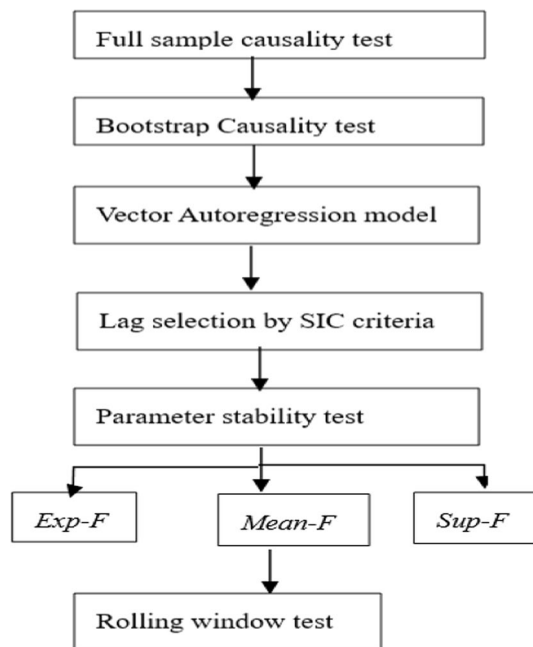


Fig. 1 step by step procedure

declined BDI. Baltic Exchange, the world's premier trade explainer, uses BDI to quantify shifts in freight rates for essential commodities like coal, iron ore, and others. BDI is retrieved from the Baltic Exchange London. Thus, BDI readily reflects the global supply and demand pattern and the economic outlook that reflects the shipping service demand [39]. It is estimated as the average of four sub-categories: Handysize, Panamax, Supramax, and Cape-size (Baltic Exchange, 2017). The monthly data is directly downloaded from the Baltic Exchange website. The data

are carefully checked for consistency and anomalies and ensured that all variables are aligned at a consistent monthly frequency. The West Texas Intermediate (WTI) spot price is employed to signify OP since its active contract in the market plays a leading role in OP fluctuations and is attained by the US Energy Information Administration (Su et al., 2017a). Similarly, global maritime trade is equivalent to the goods loaded worldwide in billions of tons and obtained from the United Nations Conference on Trade and Development (UNCTAD).

Figure 2 indicates several ups and downs of the BDI and OP due to numerous economic and geopolitical events. OP spiked due to uncertainty caused by the war against terrorism and global trade declined dramatically, which dropped the shipping service demand. The slowing economic growth and geopolitical uncertainty are reflected in the falling BDI during 2001–2002. However, the global economy recovered in 2003, increasing oil supply and maritime transportation. As a result, OP rose and touched \$35 in January 2004, further growing shipping freight [39]. In October 2004, OP reached \$50, mainly caused by the unprecedented global trade expansion. Global trade growth is twice the previous year's, especially in emerging countries, which has boosted BDI. The most important is the extraordinary increase in Chinese imports, which raises production and OP. Meanwhile, the Organization of Petroleum Exporting Countries (OPEC) alters the overall price range mechanism, increasing OP. However, due to China's restrictions on importing iron ore by domestic steel companies, the demand for shipping services has fallen, and annual maritime trade declined by 1.5% in 2005. BDI rose and reached its highest point in November 2007, mainly because of strong

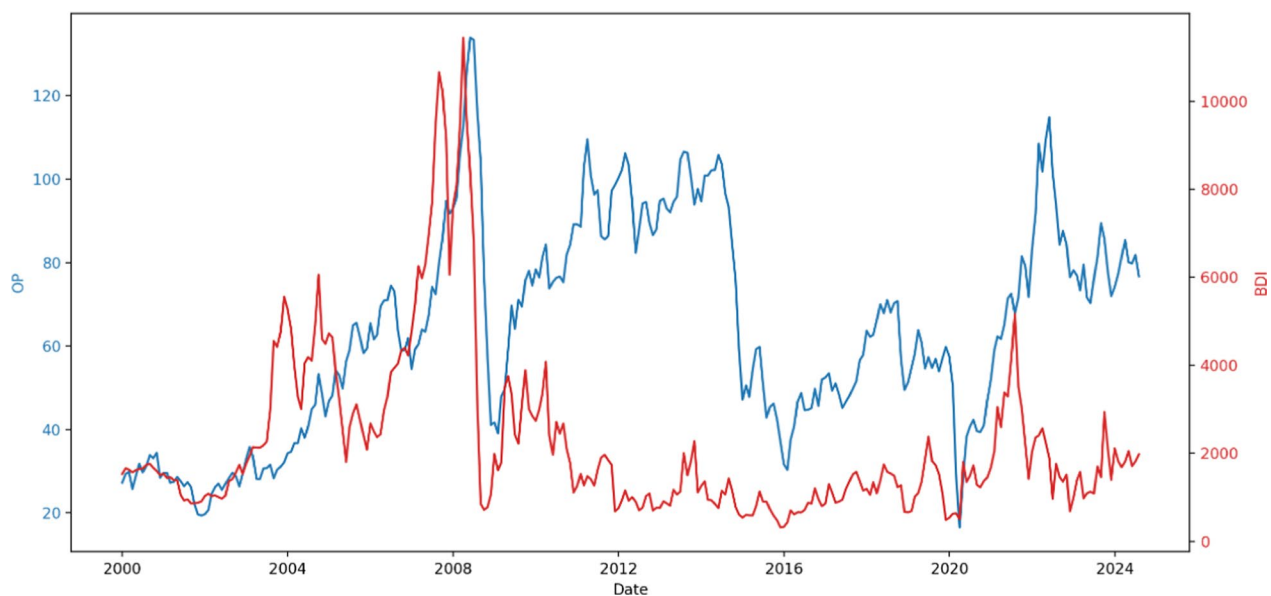


Fig. 2 Graphic presentation of OP and BDI

economic growth in China and India. Moreover, the global dry bulk trade of steel increased due to the Chinese metal industries' demand. OP rises to the highest level during the same period because of the geopolitical situation and inadequate oil reserves. However, OP and BDI declined simultaneously in 2008 due to the global financial recession. BDI reached its highest level in 2008 and dropped sharply in December 2008. The financial crisis has caused BDI and OP to fall by 90 percent and 80 percent [42]. Weak economic activity and international trade contributed to decreased oil consumption during the crisis [19]. Similarly, the marine traffic of iron ore, coal, steel, and other commodities reduces dry trade [39].

In the post-recession period, OP increased in 2011 due to higher demand and stable economic growth. The most significant concern is slowing China's growth which is the largest oil consumer. However, the OPEC and the US oil shale supply remarkably increased in 2011–2013, which is responsible for the large swing of OP. Thus, the oversupply of oil is a vital factor behind the OP decline onward 2014. However, BDI declined by 50% in 2014, which recovered in the first half of 2015 due to a positive outlook and stabilization in Europe. In 2016, OP remained volatile and low, mainly caused by shale oil production and increasing OPEC supply. Similarly, BDI reached the lowest level of 290 in 2016 (Bloomberg, 2016; [25]). OP collapses due to the low demand and the economic activity halt, followed by widespread lockdowns and social isolation. BDI plunges deep because of coronavirus and uncertainty around its effect on worldwide trade, trailed by bottommost commodity rates. The oil prices collapsed in mid-March 2020, which contracted the international maritime trade and the major freight rates witnessed their lowest level. BDI is significantly impacted during the first half of 2020, mostly due to global trade since it is the expense of shipping dry bulk materials. BDI shows a mild upward trend in 2020 which can be explained by the fact that some Asian countries have resumed their economic activities due to a receding number of new cases. It has increased demand for the goods, leading to shipping demand in late 2020. However, the shipping industry endures the pandemic until the first quarter of 2021, and the situation is reversed in the second quarter of 2021. BDI moves upward rapidly as the economies are resuming economic activities and increasing the raw-materials demand, increasing the freight plea and putting pressure on the freight rates. Moreover, the stranded ships and containers have adversely put pressure on the shipping cost because of the unavailability of the containers. The shortage has exacerbated the shipping cost and has shown a historically high level. However, the shipping freights started declining in November 2021 due to the stability in the supply chains, which may help reduce the cost of the commodity.

Table 1 Summary statistics. Sources: BDI is retrieved from the Baltic Exchange London. OP is obtained from the US Energy Information Administration and GMT is sourced from UNCTAD

	OP	BDI	GMT
Mean	4.013	7.431	8.514
Maximum	4.896	9.345	11.252
Minimum	2.806	5.748	5.983
Std. Dev	0.460	0.740	1.659
Skewness	− 0.316	0.480	− 0.034
Kurtosis	2.187	2.686	1.762
J-B	10.896***	10.451***	16.125***

OP is oil price, BDI is Baltic Dry Index, GMT denotes global maritime trade

Table 2 Unit root test

	ADF	PP	KPSS
OP	− 1.442	− 1.607	1.190***
BDI	− 2.832	− 3.0344	0.636***
GMT	− 2.448	− 2.860	1.248***
Δ OP	− 11.837***	− 12.104***	0.063
Δ BDI	− 14.747***	− 14.966***	0.074
Δ GMT	− 20.266***	− 22.646***	0.071

***p = 1% and lag length as per SIC method. The figures in parenthesis express lag length according to the SIC selection

Table 1 highlights the summary statistics. OP and global maritime trade are skewed to the left, while BDI is skewed to the right. The values of the kurtosis indicate that variables are platykurtic distribution. OP and BDI experience structural changes during the period, resulting in higher uncertainty. The Jarque Bera test confirms the non-normal distribution for all series. Hence, the traditional approaches are not appropriate for detecting the causal connection between OP and BDI, which is substituted using the rolling window approach test. It determines the consistency and accuracy of the results between the two series.

5 Empirical results

The stationarity of the series is examined through the Kwiatkowski-Phillips-Schmidt-Shin [22], Phillips-Perron [32] and Augmented Dickey-Fuller [13] tests. The unit root test results are shown in Table 2. The ADF and PP have a null hypothesis that time series has a unit root. In case the null hypothesis is accepted, the series has unit root or non-stationary. On the other hand, KPSS tests the null hypothesis that the time series is stationary against the alternative hypothesis of not stationary. If the null hypothesis is rejected, it suggests that the series is non-stationary. The results suggest that it accepts the hypothesis of unit root at the level, suggesting that these variables are non-stationary. However, these are stationary at the first difference. Similarly, the OP and BDI order of integration is I(1), indicating that they become stationary after differencing.

Table 3 displays the results of a Granger causality test performed on the full sample. We reject the underlying hypothesis that OP does not cause BDI at 1% significance levels, respectively. This means that oil is the largest freight cost, and the increase in OP can be reflected in higher BDI. Similarly, BDI is important for oil demand–supply because it accounts for more than 80% of the international commodity market.

It is assumed that there is no structural transformation and contains one causality in the whole sample. However, the statement does not establish causation between OP and BDI, highlighting the structural changes produced by various policy shifts and political and economic catastrophes. Furthermore, the existence of shocks results in short-term parametric instability, which renders the result invalid. Thus, the parametric stability tests measure the model's short-term consistency through *Exp-F*, *Mean-F* and *Sup-F*, as reported in Table 4. Therefore, the null hypothesis is rejected for both OP and BDI, implying that both equations have short-term instability. Instead, the *Exp-F* and *Mean-F* are applied to verify parameter variations and the hypothesis is rejected, suggesting that OP and BDI are evolving. However, the *Lc* test is utilized to investigate the overall VAR system. It tests the parameter constancy and the finding explores a random walk, thus establishing that the full VAR model has short-term variability. Therefore, it can be concluded that as a result of structural changes, the overall outcome faces the complication of parameter constancy in the short term, and the entire sample is unreliable.

Bai and Perron's (1998) test is employed to detect structural issues and examine structural breaks in OP and BDI. The test allows multiple breaks and determines the breakpoint of the variables. The results are shown in Table 5. It shows that these variables experience several structural breaks.

The rolling window is utilized to consider the structural changes and find a time-varying nexus between OP and BDI. The time-varying properties of the sub-sample distinguish the traditional and rolling window approaches. The underlying hypothesis that OP does not cause BDI is evaluated, and the VAR model produces the bootstrap *p*-values. Figure 3 depicts the findings that the null hypothesis is rejected across many sub-samples, spanning 2004:01 to 2004:07, 2006:06 to 2007:02, 2008:08 to

Table 3 Full sample Granger causality test

Test i	H ₀ : OP does not Granger cause BDI		H ₀ : BDI does not Granger cause OP	
	Statistics	<i>p</i> -values	Statistics	<i>p</i> -values
Bootstrap	11.589*	0.003	4.740	0.122
LR test i				

p* = 10%, **p* = 1%

Table 5 Bai-Perron test of L + 1 vs. L sequentially determined breaks

Break tests	OP		BDI	
	<i>F</i> -statistics	Critical value	<i>F</i> -statistics	Critical value
0 vs 1*	581.558***	8.58	85.196***	8.58
1 vs 2*	68.398***	10.13	110.992***	10.13
2 vs 3*	157.686***	11.14	21.878***	11.14
3 vs 4*	62.984***	11.83	47.163***	11.83
Break date		2005M06		2003M09
		2010M12		2008M08
		2014M12		2012M04
		2021M01		2021M01

***denotes significance at the 1% level

2010:11, and 2017:07 to 2018:07. It demonstrates when OP fell as a result of slowing down commercial activities, and raw material demand was reduced, which led to a decrease in BDI [42].

On the other hand, Fig. 4 highlights the results confirming the causal relation concerning OP and BDI. It shows positive connections between OP and BDI across the sub-sample consisting of 2004:01–2004:07, 2008:08–2010:11, and 2017:07–2018:07. It shows that from 2004:01 to 2004:07, both OP and BDI expand in the same direction. Figure 4 highlights the time-varying properties. OP started to rise in 2004 because of global economic improvement, and the growth rate reached an average of 2.6%, increasing shipping service demand. More specifically, the extraordinary economic performance of China, as well as the world [39], improves commodities demand such as iron ore, coal, and copper. Similarly, global maritime trade has increased, mainly dominated by the extraordinary performance of Chinese imports. On the other hand, OP is showing an upward trend, characterized by huge demand and fear of supply disruption due to geopolitical tensions and speculation, and has a substantial effect on BDI [10]. OP records a high level of \$64.5 and may negatively impact economic

Table 4 The parameter stability test

	OP equation		BDI equation		VAR system	
	Statistics	<i>p</i> -values	Statistics	<i>p</i> -values	Statistics	<i>p</i> -values
<i>Sup-F</i>	40.418***	0.000	53.111***	0.000	82.931***	0.000
<i>Mean-F</i>	19.410***	0.002	17.289***	0.020	28.072***	0.001
<i>Exp-F</i>	16.178***	0.000	22.223***	0.000	35.654***	0.000
<i>Lc</i>					3.344**	0.005

**p* = 10% and *p* = 5%

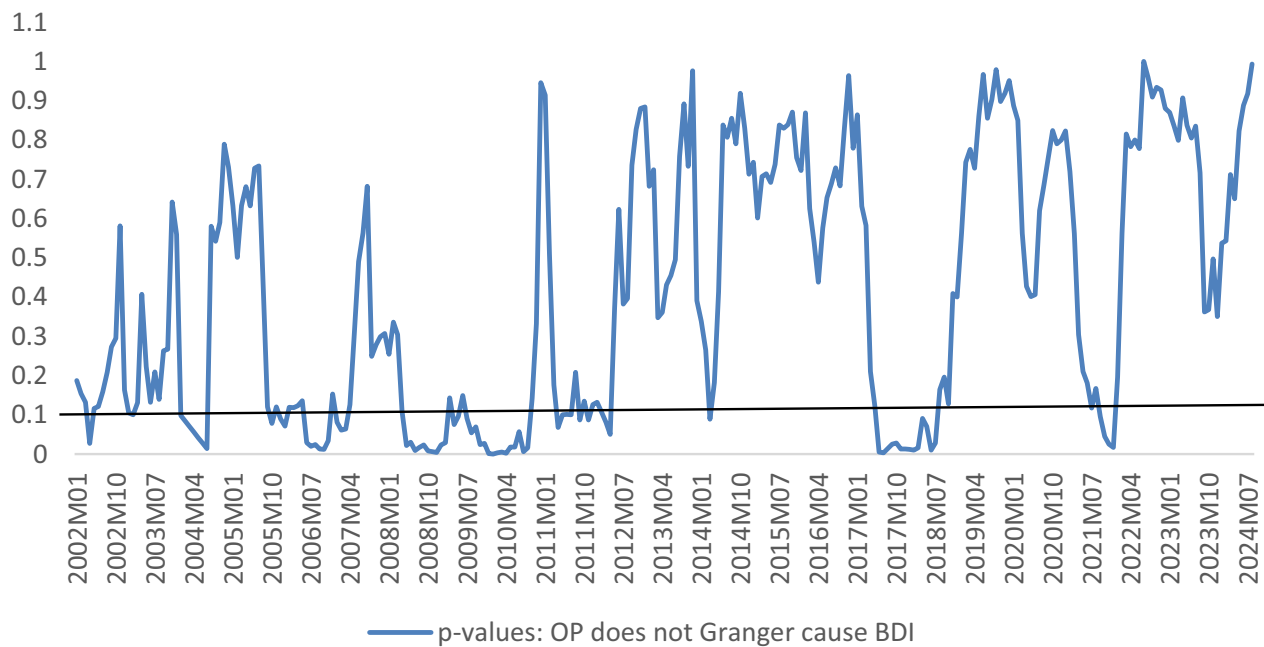


Fig. 3 OP causal impact on BDI

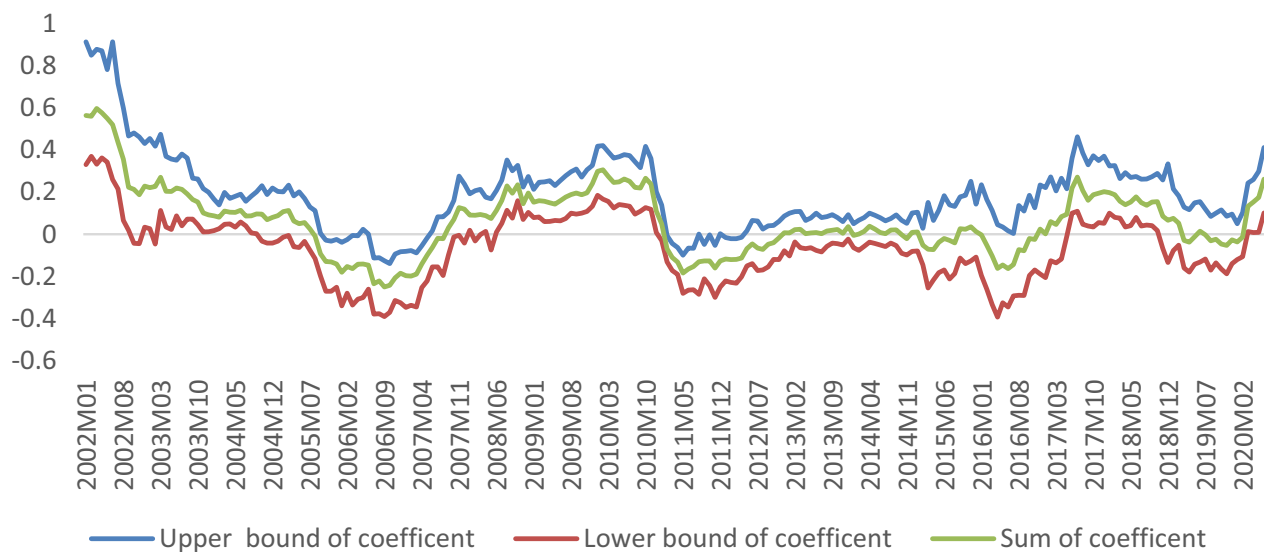


Fig. 4 Trend of OP impact on BDI

activity. It makes sense that higher OP generally reflects a shortage of shipping capacity against demand. Economic growth motivates maritime trade and increases the need for shipping services. However, OP is one of the considerable costs for shipping that pushes BDI [15]. Similarly, sub-samples of 2006:06 to 2007:02 indicate the negative nexus between OP and BDI. During this time, the world economic growth rises at 5% while the oil supply is low, reducing consumption and increasing OP. The gap between demand and supply is the main reason for the high OP. The huge demand from emerging economies exerts pressure on OP while the producer cannot meet

the surging demand. The uncertainty about the future availability of production and supply of oil may take longer to adjust against the demand [19]. The OPEC countries demand a price rise in oil to balance their budget. Consequently, OPEC cut production in 2007 to seek a price increase. The prevailing uncertainty and vacillating economy cause OPEC to keep OP high. Similarly, the imbalance between demand and supply results in the rising OP, and the shipping freight has seen an upward trend. However, beyond 2006, the slowing down of the US economy and uncertainty about China's outlook contributed to the lower BDI.

From 2008:07 to 2010:07, OP is positively leading BDI, which means that both moved in the same direction. The demand for raw materials is increased due to higher economic growth, which boosts maritime transportation costs [8]. With the rising economic growth, OP peaked in 2008, increasing the freight rates. However, recession turns the global economy, the slowing economic activity results in the decline of OP and commodity demand. This has also reduced the demand for the shipping industry, and its earnings declined by 80%. Likewise, the financial crisis declines BDI and OP by more than 90% and 80%, respectively [42]. Also, BDI recovers simultaneously with OP in the post-recession period determining economic recovery in the major economies. However, OP rises more quickly compared to BDI in the post-financial scenario. It might be linked to the global incidence of uncertainty. The weak economic revival considerably influences shipping demand, also slowing recovery. Likewise, OP positively impacted BDI during 2017:07–2018:07, similar to Alizadeh and Nomikos [3], who confirm the connection between OP and BDI. During the period, OP started rising due to demand, especially from China, because of its remarkable imports, which resulted in an upward trend in BDI [39, 41]. However, the US and China trade conflict started in 2018, which has a huge economic cost for both countries. This is a catastrophe for the international trade and commodity markets, which are declining, as shown by declining BDI [23].

Figure 5 reveals causality from BDI to OP across sub-samples such as 2010:05–2010:12, 2014:01–2014:06, 2016:05–2016:09, and 2017:04–2017:07. The hypothesis that BDI causes OP is rejected at a 10% significance level.

Instead, it implies that BDI is one of the predictors of OP; transportation costs are essential as fuel is the biggest operating expense for the shipping industry and can play a significant in the commodity market.

However, the finding also reveals the direction of the causality between BDI and OP, illustrated in Fig. 6. According to the results, BDI Granger causes OP positively in the following sub-samples; 2010:05–2010:12, 2014:01–2014:06; 2016:05–2016:09, 2017:04–2017:07. The outcome records a positive relationship between BDI and OP from 2010:05 to 2010:12. The period is characterized as the recovery phase and both BDI as well as OP show a rising pattern because of growing economic development. Similarly, supply interruption owing to geopolitical tensions in key producing nations such as Iraq, Russia, Iran, and Libya are a concern [41]. This has increased OP, and the same trend is reflected in BDI. Similarly, BDI positively impacted OP during 2014:01–2014:06. The vertical fall was witnessed in OP during 2014, influencing the shipping and overall economic activity. Due to the economic slowdown in 2014, China has lowered its oil and raw materials demand, adversely affecting the shipping industry and BDI [23]. It declined in 2014 and recovered in the first half of 2015 due to a positive outlook and stabilization in Europe. The sharp decline in OP in 2014 indicates a lower demand for products and commodities, resulting in a falling BDI [24]. Similarly, during 2016:05–2016:09, BDI and OP moved on a similar line. In 2016, OP remained volatile and low, mainly triggered by shale oil production and increasing OPEC supply. Likewise, BDI touched its lowest level in 2016 [25]. Lastly, the period from 2017:04 to 2017:07

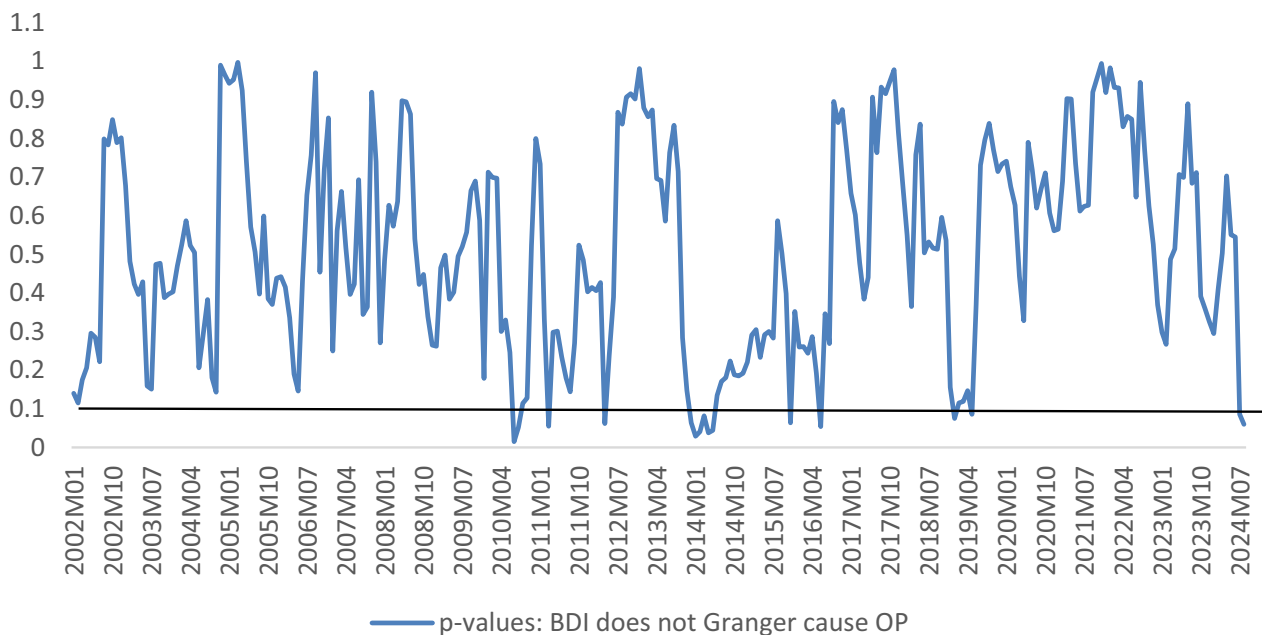


Fig. 5 BDI causal impact on OP

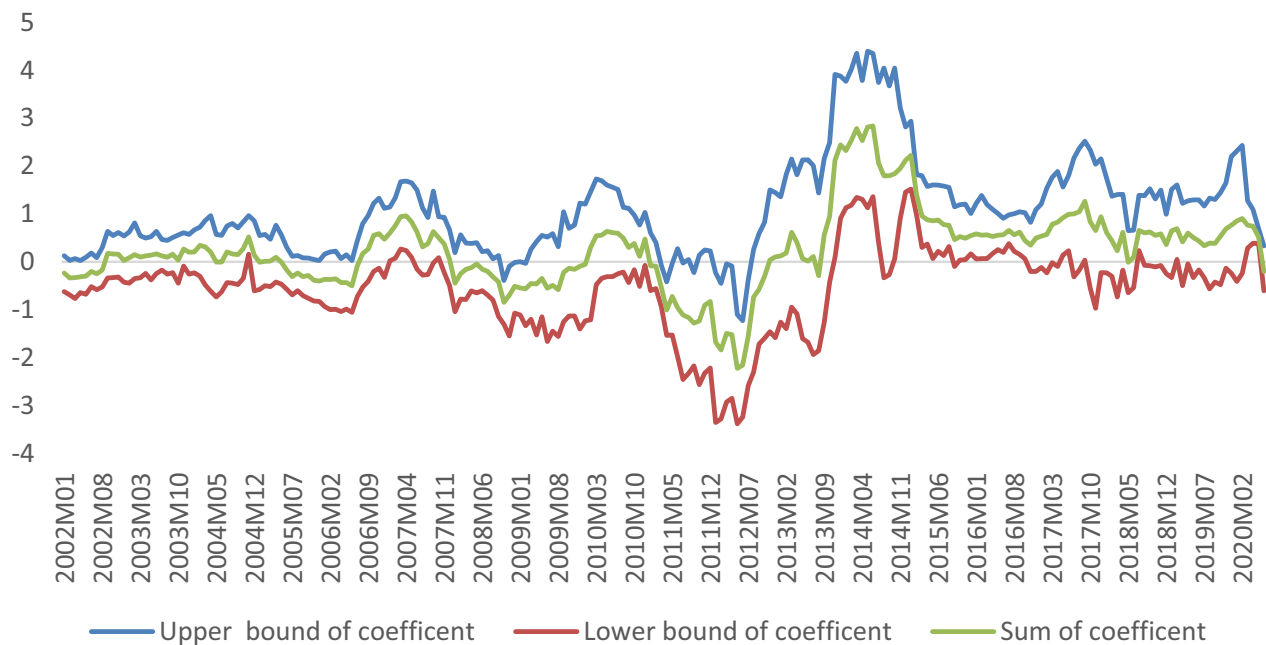


Fig. 6 Direction of BDI impact on OP

expresses the positive effect of BDI on OP, which is attributed to economic stability and higher demand for raw materials from emerging economies. This has boosted the need for shipping services and OP [39].

6 Discussion and robustness test

6.1 Discussion

The result summary shows that OP plays an essential role in BDI because fuel is one of the biggest operating expenses [36, 40]. Moreover, the result displays the bi-directional causality between OP and BDI [20]. Consequently, the rolling-window process is employed to inspect the interconnection between OP and BDI across the sub-samples. The outcomes explore the positive impact of OP on BDI and vice versa in different sub-samples, particularly coinciding with the political or economic event. Similarly, the time-varying nature is observed in both Figs. 3 and 5. The study found that OP significantly impacts BDI, primarily due to global economic growth, variable oil supply, OPEC regulations, global marine commerce, the 2008 financial crisis, and the US-China confrontation. Similarly, the BDI greatly influences OP due to rising economic growth, supply interruption due to geopolitical tensions in key producing nations, China cutting demand, and the 2014 oil crash. The period causality shows in common that the relationship coincides with the major economic and political events. Therefore, the knowledge of potential OP may lessen the risk of abrupt changes in BDI. Furthermore, information on various sources of OP fluctuations can help the shipping industry take practical measures to

minimize the impact of shocks. Therefore, the stability of OP may depend on the smooth operation of the shipping industry.

6.2 Robustness test

The robustness test is conducted to make the main results valid and convincing. For this purpose, the wavelet analysis is employed to examine OP and BDI correlation across different sub-samples. The results have shown (See Fig. 7) that OP and BDI have a lead lag relationship during 2006–2007; 2008–2010; 2014–2016 and 2017–2018 influenced by global economic growth, fluctuating oil supply, OPEC policies, international maritime trade, the 2008 financial crisis, and the U.S.-China conflict. Similarly, the BDI significantly impacts oil prices across various subsamples. The causality is attributed to economic growth, supply disruptions due to geopolitical tensions, reduced demand from China, and the oil market collapse. The significance of this causal relationship underscores its alignment with major economic and political events. The results support the main results of the rolling window.

7 Conclusion

The paper examines the causality between OP and BDI across the sub-samples through the rolling window procedure. The results indicate the causal influence of OP on BDI and vice versa in various sub-samples, especially coinciding with political or economic events, which can significantly impact world economic growth and OP supply and demand. Moreover, OP is causing BDI for a longer

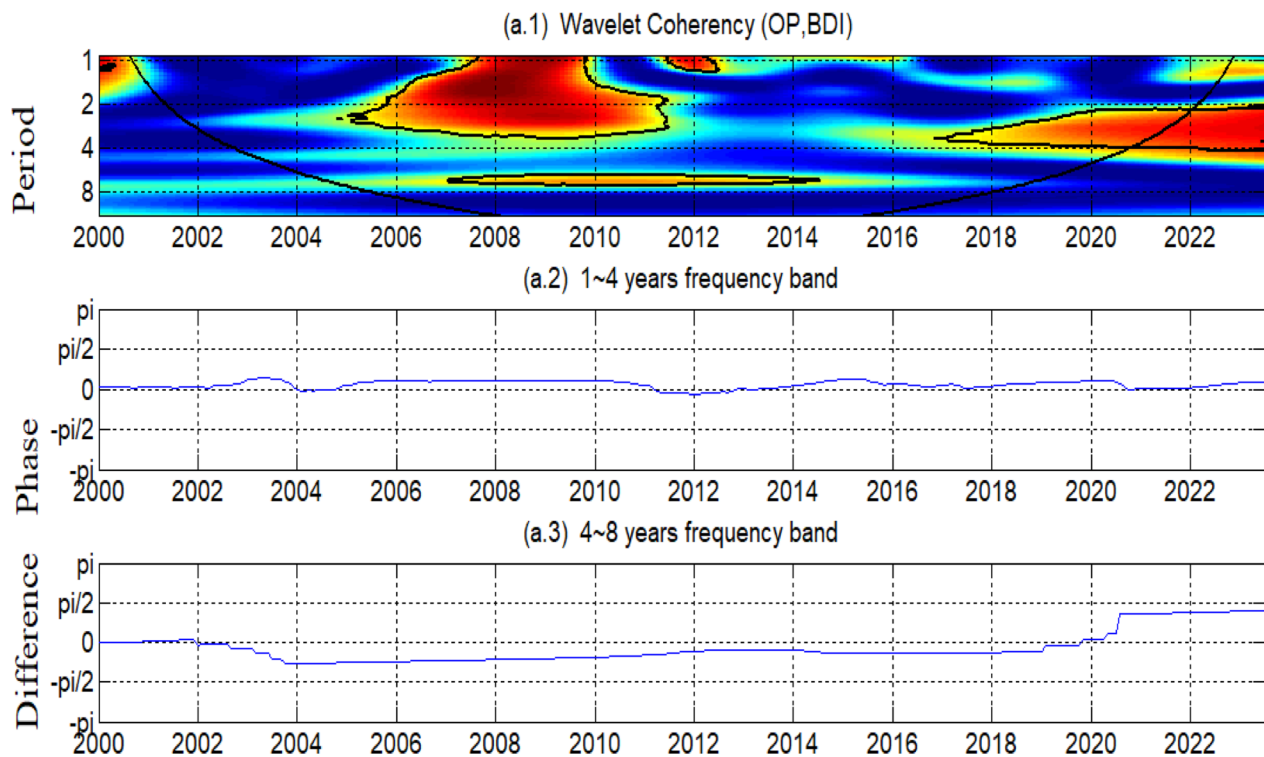


Fig. 7 Wavelet coherence and phase difference between OP and BDI

period than BDI, suggesting that OP has a more contributing role in BDI. It implies that OP is the main energy source of the shipping and the oil is carried through the shipping. Thus, the volatility in OP may translate into the shipping demand. The rise in fuel prices can quickly reverberate in freight, whereas BDI changes can decrease shipping services demand, resulting in lower supply and eventually affecting OP. The finding explores that OP has a significant effect on BDI across several sub-samples; including 2004:01 to 2004:07, 2006:06 to 2007:02; 2008:08 to 2010:11; 2017:07 to 2018:07, mainly caused by the global economic growth, fluctuating oil supply, OPEC policies, international maritime trade, the financial crisis in 2008 and the U.S. China conflict. Similarly, the BDI has a significant impact on OP across subsamples such as 2010:05–2010:12, 2014:01–2014:06, 2016:05–2016:09, and 2017:04–2017:07. The causality is caused by growing economic development and supply disruption because of geopolitical tension in the main producing nations such as Iraq, Russia, Iran and Libya, China lowering demand and oil collapsed in 2014.

The study presents both theoretical and practical insights. Theoretically, it is one of the few contributions in the field of OP and BDI that expands upon contemporary literature. It demonstrates how the relationship between OP and BDI evolves. On the other hand, several policy implications can be derived from the study as follows. First, the findings indicate that OP plays a

more dominant role in the formation of the BDI due to oil being one of the major operating costs. The demand for shipping may be influenced by fluctuations in oil demand. Therefore, understanding potential changes in OP could help mitigate the risk of sudden spikes in the BDI, which can negatively impact shipping demand and global trade. A clear understanding of the various sources of OP fluctuations enables the shipping industry to implement practical measures to minimize the effects of these shocks. This could involve establishing a centralized data repository to analyze OP trends and incorporating this information into operational planning, allowing shipping companies to adopt risk management strategies such as hedging or diversifying fuel sources. Additionally, establishing partnerships with energy analysts and market experts can enhance the industry's preparedness for oil market shifts. Furthermore, the study offers valuable insights for stakeholders in the shipping industry, emphasizing the importance of understanding the behavior of the OP and BDI to ensure stability and smooth functioning within the global economy. Abnormal fluctuations in supply or demand can adversely affect international trade, making it essential to implement timely measures to prevent unforeseen disruptions and maintain stability. Second, the finding reveals that BDI causes OP in the short run, highlighting the importance of the shipping industry as most of the trade is carried through the ocean. BDI is a leading indicator of global demand and supply;

business conditions and economic growth push shipping service demand. Therefore, the shipping industry can be critical in the international oil market. Policymakers should encourage investment in shipping infrastructure and technology, enhancing industry response to demand and supply fluctuations, and establishing innovation and operational efficiency, stabilizing the oil market. The study provides valuable insights for shipping industry stakeholders regarding the specific periods and factors influencing OP and BDI dynamics. To maintain stability in international trade, stakeholders must establish an early warning system that monitors key economic indicators and geopolitical events affecting oil supply and shipping demand. Last, in terms of the methodology, the rolling window is used, which examines the time-varying properties while considering structural changes. The rolling window has the distinction of analyzing the causality at the subsample level with high power to detect the time variation. Thus, the method precisely examines the causality between OP and BDI in different sub-samples. The study observed numerous political and economic events, such as the global financial crisis, eurozone debt, the collapse of oil in 2014, and COVID-19. Thereby, the model allowed the time variation characteristic and investigated the causality. The study can examine the impact of OP on different shipping freights considering the exogenous and dummy variables, which can explore the role of these macroeconomic variables on shipping freights. Moreover, frequent fluctuations in shipping freight rates, necessitate a study to understand the reasons behind these fluctuations and the specific coinciding period, aiding policymakers. In the event that long-term hedges fail, shipping companies operating in energy-intensive industries can use oil price hedging options to reduce financial risk and stabilize pricing. To mitigate additional regional disruptions through route diversification and fleet management, these include long-term fixed pricing contracts with fuel surcharges to provide some protection against price and shipping rate uncertainty. To promote energy security and fuel price stabilization, the main goal is to emphasize the necessity of considering the intricate relationships between energy prices and global supply chain costs when developing energy and transportation policies.

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